FINITELY GENERATED MODULES ARE EXACTLY THOSE FOR WHICH TENSOR COMMUTES WITH DIRECT PRODUCTS SURJECTIVELY

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Editorial note. The counted proof contains an Xy-pic diagram; the preamble spans of this card run 8-18 of the chapter's own preamble file and therefore include the Xy-pic package and option setup at lines 15-18.

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Allowed standard prerequisites (editorial).

- Tensor products commute with finite direct sums. Used for the top arrow of the counted diagram (lines 21551-21557). Makes the top horizontal map an isomorphism for a finite free module, which is what the diagram chase consumes.
- Right exactness of the tensor product. Used at proof lines 21553-21558. Turns the surjection from a finite free module onto M into a surjection between the tensored products.

Proposition 0.1. *Let M be an R-module. The following are equivalent:*

(1) *M is finitely generated.*
(2) *For every family $(Q_\alpha)_{\alpha\in A}$ of R-modules, the canonical map $M\otimes_R(\prod_\alpha Q_\alpha) \to \prod_\alpha(M\otimes_R Q_\alpha)$ is surjective.*
(3) *For every R-module Q and every set A, the canonical map $M\otimes_R Q^A \to (M\otimes_R Q)^A$ is surjective.*

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(4) *For every set A, the canonical map $M \otimes_R R^A \to M^A$ is surjective.*

Proof. First we prove (1) implies (2). Choose a surjection $R^n \to M$ and consider the commutative diagram

$$\begin{array}{ccc} R^n \otimes_R (\prod_\alpha Q_\alpha) & \xrightarrow{\cong} & \prod_\alpha (R^n \otimes_R Q_\alpha) \\ \downarrow & & \downarrow \\ M \otimes_R (\prod_\alpha Q_\alpha) & \longrightarrow & \prod_\alpha (M \otimes_R Q_\alpha). \end{array}$$

The top arrow is an isomorphism and the vertical arrows are surjections. We conclude that the bottom arrow is a surjection.

Obviously (2) implies (3) implies (4), so it remains to prove (4) implies (1). In fact for (1) to hold it suffices that the element $d = (x)_{x \in M}$ of M^M is in the image of the map $f : M \otimes_R R^M \to M^M$. In this case $d = \sum_{i=1}^n f(x_i \otimes a_i)$ for some $x_i \in M$ and $a_i \in R^M$. If for $x \in M$ we write $p_x : M^M \to M$ for the projection onto the x-th factor, then

$$x = p_x(d) = \sum\nolimits_{i=1}^n p_x(f(x_i \otimes a_i)) = \sum\nolimits_{i=1}^n p_x(a_i) x_i.$$

Thus $x_1, \ldots, x_n$ generate M. □